

Experimental Design - Practice Multiple Choice Questions

Exam 1 Preparation: CRD, RCBD, and Factorial Designs

AGST 50104 - Spring 2026

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Instructions

- Each question has **4 options**
 - Only **one option is correct** (marked in bold in this answer key)
 - A brief explanation follows each question
 - Read experiment descriptions carefully for degrees of freedom calculations
 - Focus on identifying true vs. pseudo-replication scenarios
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Question 1

A researcher tests 5 nitrogen fertilizer rates on wheat yield using 8 replicate plots per rate in a completely randomized design (CRD). What are the degrees of freedom for **error** in the ANOVA table?

- a) 36
- b) 7
- c) **35**
- d) 39

Explanation: In a CRD, $df_{\text{Error}} = N - t = t(r - 1)$ where t = number of treatments and r = replicates per treatment. Here, $t = 5$ and $r = 8$, so $df_{\text{Error}} = 5(8 - 1) = 5 \times 7 = 35$. Alternatively, total DF = 39 and treatment DF = 4, so $39 - 4 = 35$.

Question 2

An agronomist conducts a field trial with 4 treatments arranged in 6 complete blocks. Each block contains all 4 treatments randomized within the block. What are the degrees of freedom for **treatments** in the RCBD ANOVA?

- a) **3**
- b) 5
- c) 15
- d) 23

Explanation: In RCBD, $df_{\text{Treatment}} = t - 1$ where t = number of treatments. Here, $t = 4$, so $df_{\text{Treatment}} = 4 - 1 = 3$. This is the same formula as CRD; the blocking structure doesn't change treatment df.

Question 3

A greenhouse study tests the effect of 3 irrigation levels on tomato plants. The researcher places 5 plants in each of 4 pots per irrigation level. At harvest, all 20 tomatoes per pot are weighed. What is the correct **experimental unit** in this study?

- a) Individual tomato fruit
- b) Individual tomato plant
- c) **Pot**
- d) Irrigation level

Explanation: The experimental unit is the smallest unit to which the treatment is **independently applied**. Here, irrigation is applied at the **pot level** (each pot gets one irrigation level), not to individual plants or fruits. Plants within the same pot share the same treatment application and are not independent. Therefore, **pot** is the experimental unit, and measurements on multiple plants/fruits within a pot represent **pseudo-replication** (technical replication).

Question 4

In the greenhouse study described in Question 3, the researcher reports “ $n = 20$ tomatoes per treatment” and runs an ANOVA treating each tomato as an independent replicate. What is the **problem** with this analysis?

- a) Sample size is too small
- b) ANOVA cannot be used for plant data
- c) **Pseudo-replication: multiple tomatoes per pot are not independent**
- d) Irrigation levels should be blocked

Explanation: This is a classic case of **pseudo-replication**. The true replication is at the **pot level** (4 pots per irrigation level = 4 true replicates). The 20 tomatoes within each pot are **subsamples** from the same experimental unit and violate the independence assumption. The correct analysis should use pot means or account for the nested structure (mixed model). Treating all 20 tomatoes as independent inflates the sample size artificially and leads to invalid p-values.

Question 5

A livestock nutritionist tests 4 diets on cattle. She has 24 animals and randomly assigns 6 animals to each diet, with each animal housed in a separate pen. Which experimental design is this?

- a) Factorial design
- b) **Completely Randomized Design (CRD)**
- c) Randomized Complete Block Design (RCBD)
- d) Split-plot design

Explanation: This is a **CRD** because treatments (diets) are assigned **completely at random** to all experimental units (individual animals) without any blocking structure. Each animal is an independent experimental unit in a separate pen. There is only one factor (diet) tested, so it's not a factorial design.

Question 6

An orchard experiment compares 5 apple tree varieties. The orchard has a known fertility gradient from north to south. The researcher divides the orchard into 5 strips (north to south) and plants one tree of each variety randomly within each strip. What design is this?

- a) Completely Randomized Design (CRD)
- b) **Randomized Complete Block Design (RCBD)**
- c) Latin Square Design
- d) Factorial design

Explanation: This is an **RCBD**. The strips serve as **blocks** to control the known north-south fertility gradient. Each block (strip) contains one complete set of all 5 varieties, and varieties are randomized **within each block**. This design reduces experimental error by removing block-to-block variation and allows fair comparison of varieties within similar fertility conditions.

Question 7

In an RCBD with 4 blocks and 5 treatments, what are the degrees of freedom for **error**?

- a) 4
- b) **12**
- c) 16
- d) 19

Explanation: In RCBD, $df_Error = (b - 1)(t - 1)$ where b = number of blocks and t = number of treatments. Here, $b = 4$ and $t = 5$, so $df_Error = (4 - 1)(5 - 1) = 3 \times 4 = 12$. This represents the “interaction” between blocks and treatments, which is used as the error term in RCBD.

Question 8

A factorial experiment tests 3 levels of nitrogen (0, 50, 100 kg/ha) and 2 levels of phosphorus (0, 30 kg/ha) with 4 replicates per combination in a CRD structure. What are the degrees of freedom for **error** in the ANOVA?

- a) 5
- b) 12
- c) **18**
- d) 23

Explanation: For a two-factor factorial in CRD, $df_Error = ab(r - 1)$ where a and b are the levels of each factor and r = replicates. Here, $a = 3$ nitrogen levels, $b = 2$ phosphorus levels, $r = 4$ replicates, so $df_Error = 3 \times 2 \times (4 - 1) = 6 \times 3 = 18$. Alternatively, there are 6 treatment combinations with total $N = 24$, so $df_Error = 24 - 6 = 18$. Or using df partitioning: Total DF = 23, $df_Nitrogen = 2$, $df_Phosphorus = 1$, $df_Interaction = 2$, leaving $23 - 2 - 1 - 2 = 18$ for error.

Question 9

A researcher runs a Q-Q plot of residuals from an ANOVA and observes that points form an S-curve (systematic deviation from the diagonal line). What does this indicate?

- a) Homogeneity of variance is violated
- b) **Residuals are skewed (non-normal)**
- c) Independence assumption is violated
- d) The model fits perfectly

Explanation: An S-curve pattern in a Q-Q plot indicates **skewness** in the residuals (non-normality). Points should fall on or near the diagonal line if residuals are normally distributed. An S-curve suggests either right-skew or left-skew depending on the direction. Common fix: data transformation (log, square root) or better yet, use Generalized Linear Models (GLMs) that do not assume normality.

Question 10

Which diagnostic plot is **most useful** for detecting heteroscedasticity (unequal variances)?

- a) Q-Q plot
- b) **Residuals vs. Fitted Values plot**
- c) Histogram of residuals
- d) Cook's distance plot

Explanation: The **Residuals vs. Fitted Values plot** is ideal for detecting heteroscedasticity. If variances are equal (homoscedasticity), the scatter should be random with constant spread. A “funnel shape” (increasing or decreasing spread) indicates heteroscedasticity. The Q-Q plot checks normality, not variance homogeneity.

Question 11

An ANOVA F-test for treatments yields $p = 0.001$ (highly significant). Which statement is **correct**?

- a) All treatment means are significantly different from each other
- b) The largest mean is significantly different from the smallest mean
- c) **At least one treatment mean differs from the others**
- d) The treatment effect is practically important

Explanation: A significant F-test in ANOVA only tells us that **at least one treatment mean differs** from the others. It does **not** tell us which specific pairs differ or how many pairs differ. To determine specific pairwise differences, we need post-hoc tests (e.g., Tukey HSD). Also, statistical significance ($p < 0.05$) does not guarantee practical importance; we must examine effect sizes.

Question 12

Which post-hoc test is **most appropriate** when you want to compare all treatment means to a single **control group** only (not all pairwise comparisons)?

- a) Tukey HSD
- b) Bonferroni
- c) Fisher's LSD
- d) **Dunnett's test**

Explanation: Dunnett's test is specifically designed for comparing multiple treatment means to a **single control** group. It is more powerful than Tukey HSD in this scenario because it focuses only on control comparisons (not all pairwise). Tukey HSD is used when comparing **all pairwise** differences. Bonferroni is overly conservative. Fisher's LSD doesn't adequately control familywise error rate.

Question 13

A 2×3 factorial experiment has 2 levels of tillage (conventional, no-till) and 3 levels of cover crop (none, rye, vetch). The ANOVA shows a significant **Tillage $\times$ Cover Crop** interaction ($p = 0.003$). How should you interpret the results?

- a) Report only the main effect of tillage
- b) Report only the main effect of cover crop
- c) Report both main effects; ignore the interaction
- d) **Examine simple effects; interaction takes precedence over main effects**

Explanation: When a significant **interaction** is present, main effects alone are **misleading** because the effect of one factor **depends on** the level of the other factor. You must examine **simple effects** (e.g., effect of cover crop at each tillage level separately) and interpret the interaction using interaction plots. Reporting only main effects when interaction is significant can lead to incorrect biological conclusions.

Question 14

An ANOVA F-test yields $F = 9.0$ with $df = (5, 24)$ and $p = 0.00012$. What is the **correct interpretation** of this p-value?

- a) There is a 0.012% chance the null hypothesis is true
- b) There is a 99.988% chance the treatment effects are real
- c) **If H_0 were true, there's a 0.012% chance of observing an F-statistic this large or larger**
- d) The probability that all treatment means are equal is 0.00012

Explanation: The p-value represents the probability of observing data as extreme as (or more extreme than) what we observed, **assuming the null hypothesis is true**. It does NOT tell us the probability that H_0 is true or that the alternative is true. A p-value of 0.00012 means that if there truly were no treatment differences (H_0 true), we would observe an F-statistic ≥ 9.0 only 0.012% of the time by random chance alone. This constitutes very strong evidence against H_0 .

Question 15

A field trial with 4 treatments and 6 complete blocks (RCBD) shows $SS_{Treatment} = 240$, $SS_{Block} = 180$, $SS_{Error} = 120$. What is MS_{Error} ?

- a) 6
- b) 8
- c) 10
- d) 15

Explanation: In RCBD, $MS_{Error} = \frac{SS_{Error}}{df_{Error}}$. First, calculate $df_{Error} = (b - 1)(t - 1) = (6 - 1)(4 - 1) = 5 \times 3 = 15$. Then, $MS_{Error} = \frac{120}{15} = 8$.

Question 16

When should you use **blocking** in an experimental design?

- a) When you want to increase the number of treatments tested
- b) **When there is a known source of variation you want to control**
- c) When sample size is too small for a CRD
- d) When you need to test interactions between factors

Explanation: Blocking is used to **control known sources of variation** (e.g., spatial gradients, time differences, genetic variation) that are not of primary interest but could obscure treatment effects. By grouping experimental units into homogeneous blocks and ensuring each block contains all treatments, we remove block-to-block variation from the error term, increasing precision and statistical power. Blocking does not increase treatments, is not about sample size, and is not for testing interactions (factorial designs test interactions).

Question 17

What is the **primary benefit** of randomization in experimental design?

- a) It increases sample size

- b) It ensures equal group sizes
- c) **It minimizes bias by ensuring treatment assignment is not confounded with unknown factors**
- d) It guarantees statistically significant results

Explanation: Randomization ensures that treatment assignment is **independent of both known and unknown confounding variables**. This prevents systematic bias and allows us to make valid causal inferences. Random assignment does not increase sample size, does not require equal groups (though balanced designs are preferred), and does not guarantee significance—it simply ensures that differences observed are not due to pre-existing systematic differences between groups.

Question 18

A food scientist wants to study the effects of baking temperature and mixing time on cake texture. She considers two approaches: (1) **One-factor-at-a-time (OFAT)**: test 3 temperatures at fixed mixing time, then test 3 mixing times at fixed temperature; or (2) **Factorial design**: test all 9 combinations of 3 temperatures $\times$ 3 mixing times. What is the **main advantage** of the factorial design over OFAT?

- a) Factorial designs require more experimental units
- b) Factorial designs are easier to analyze statistically
- c) **Factorial designs can detect interactions between factors**
- d) Factorial designs always yield significant results

Explanation: The critical advantage of **factorial designs** is their ability to **detect interactions**—situations where the effect of one factor depends on the level of another factor. In the cake example, temperature might have different effects depending on mixing time (an interaction). OFAT experiments **cannot detect interactions** because factors are never varied together. Additionally, factorial designs are more efficient at estimating main effects because each factor is evaluated across all levels of the other factors. However, they are not necessarily easier to analyze or guaranteed to yield significant results.

Question 19

Fisher's three principles of experimentation are:

- a) Sampling, estimation, hypothesis testing
- b) Normality, independence, homogeneity
- c) **Replication, randomization, blocking**
- d) Power, precision, bias

Explanation: Sir Ronald Fisher established the three foundational principles of experimental design: **Replication** (multiple units per treatment to estimate variability), **Randomization** (random assignment to eliminate bias), and **Blocking** (grouping similar units to control known variation). These principles are still the cornerstone of experimental design today.

Question 20

A bakery researcher wants to optimize bread quality by testing 3 flour types and 3 yeast strains. Using a **one-factor-at-a-time (OFAT)** approach with 4 replicates per treatment, she would test 3 flour types (holding yeast constant) then 3 yeast strains (holding flour constant) = 24 total loaves. If she instead uses a **3×3 factorial design** with 4 replicates per combination, how many loaves would she bake, and what is the key efficiency advantage?

- a) **36 loaves; each main effect is estimated from all 36 observations (hidden replication), not just 12**
- b) 24 loaves; factorial designs use the same number of runs as OFAT
- c) 36 loaves; factorial designs always detect interactions
- d) 9 loaves; factorial designs are more efficient than OFAT

Explanation: In a 3×3 factorial with 4 reps, total observations = $3 \times 3 \times 4 = 36$ loaves. The key advantage is **hidden replication**: even though we have 4 true replicates per combination, each **main effect** (e.g., flour type) is estimated using **all 36 observations**. For flour type, each level appears with all 3 yeast levels (3 yeast × 4 reps = 12 obs per flour level), giving much more precision than OFAT which would use only 12 observations total for flour (4 reps × 3 levels). The effective sample size for estimating main effects is much larger in factorial designs. Additionally, factorials can detect interactions that OFAT completely misses. While factorial designs require more runs than OFAT (36 vs 24), they provide vastly more information per observation.

Bonus Question 21

An RCBD experiment with 5 treatments and 4 blocks yields the following ANOVA table:

Source	df	SS	MS	F
Block	3	120	40	-
Treatment	4	480	120	10.0
Error	12	144	12	-
Total	19	744	-	-

If we had analyzed this same data as a **CRD** (ignoring blocks), what would happen to the **error term**?

- a) Error would decrease
- b) Error would stay the same
- c) **Error would increase (SS_Error would include SS_Block)**
- d) Cannot determine from given information

Explanation: In RCBD, blocking **removes** block-to-block variation from the error term. If we ignore blocks and analyze as CRD, the block variation ($SS_{Block} = 120$) would be **absorbed into the error term**. New $SS_{Error(CRD)} = SS_{Block} + SS_{Error(RCBD)} = 120 + 144 = 264$, with $df_{Error} = 5(4 - 1) = 15$ (instead of 12 in RCBD). This leads to a larger MS_{Error} , reduced F-statistic, and lower power to detect treatment effects. This demonstrates **why blocking is beneficial** when heterogeneity exists.